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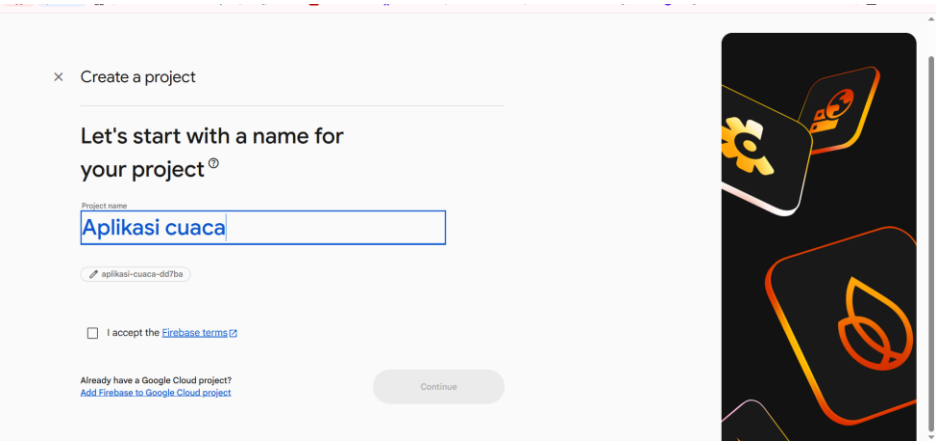
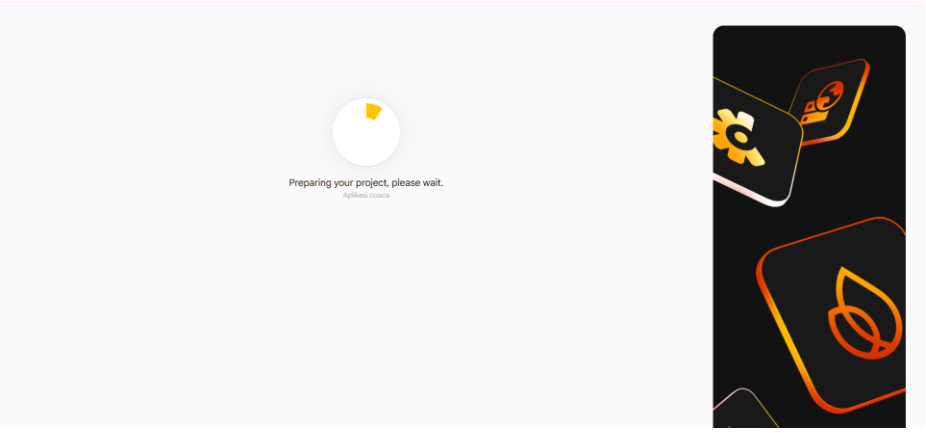
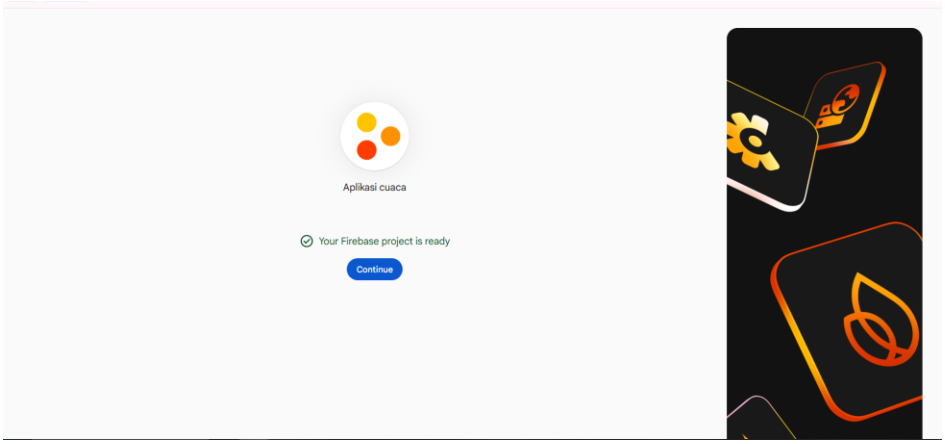
KELAS : SIB 2G
MATERI : Uji Coba Api

1. Uji coba API Firebse

- Membuat proyek baru di firebase







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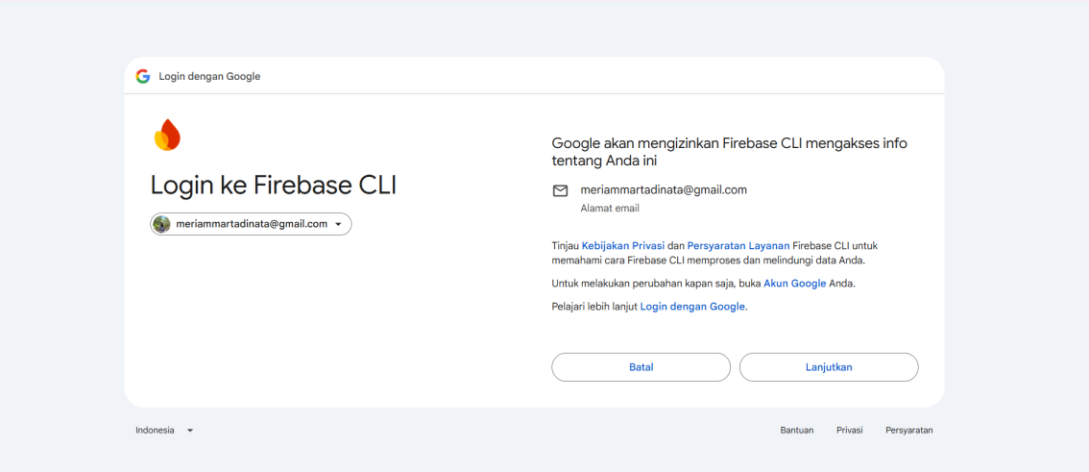
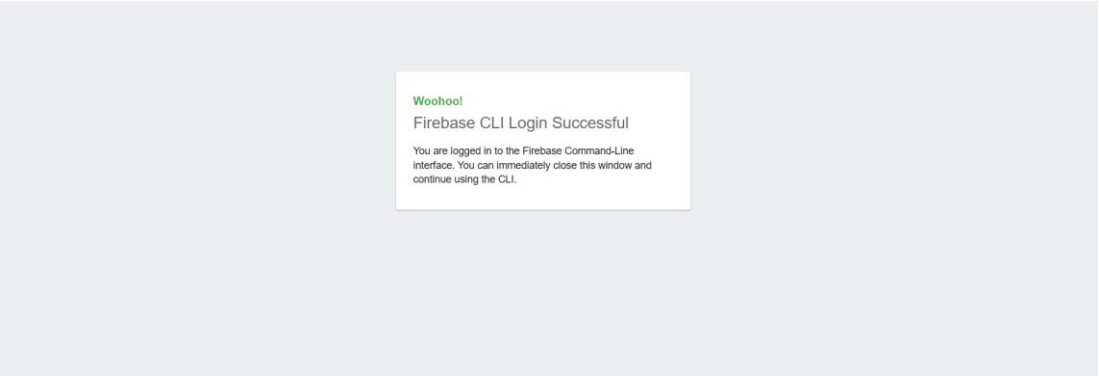
- Instalasi Firebase Tools CLI secara global menggunakan Node Package Manager (NPM).

```
PS D:\SEMESTER 4\Pemrograman-Mobile\WEEK 8-API\src> npm install -g fireba
se-tools
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'firebase-tools@15.17.0',
npm WARN EBADENGINE   required: { node: '>=20.0.0 || >=22.0.0 || >=24.0.0' },
npm WARN EBADENGINE   current: { node: 'v18.8.0', npm: '8.18.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'lsofi@2.0.0',
npm WARN EBADENGINE   required: { node: '>=20.0.0' },
npm WARN EBADENGINE   current: { node: 'v18.8.0', npm: '8.18.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'superstatic@10.0.0',
npm WARN EBADENGINE   required: { node: '20 || 22 || 24' },
npm WARN EBADENGINE   current: { node: 'v18.8.0', npm: '8.18.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
```



- Verifikasi keberhasilan login pada Firebase



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```
v24.15.0
PS D:\SEMESTER 4\Pemrograman-Mobile\WEEK 8-API\src> firebase login
i  The Firebase CLI's MCP server feature can optionally make use of Gemini in Firebase. Learn more about Gemini in Firebase and how it uses your data: https://firebase.google.com
/docs/gemini-in-firebase#how-gemini-in-firebase-uses-your-data
✓ Enable Gemini in Firebase features? Yes

i  Firebase optionally collects CLI and Emulator Suite usage and error reporting information to help improve our products. Data is collected in accordance with Google's privacy p
olicy (https://policies.google.com/privacy) and is not used to identify you.
✓ Allow Firebase to collect CLI and Emulator Suite usage and error reporting information? No

i  To change your preferences at any time, run `firebase logout` and `firebase login` again.

Visit this URL on this device to log in:
https://accounts.google.com/o/oauth2/auth?client_id=563584335869-fgrhgmd47bqnekij5i8b5pr03ho849e6.apps.googleusercontent.com&scope=email%20openid%20https%3A%2F%2Fwww.googleapis.c
om%2Fauth%2Fcloudplatformprojects.readonly%20https%3A%2F%2Fwww.googleapis.com%2Fauth%2Ffirebase%20https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform&response_type=code&stat
e=769323933&redirect_uri=http%3A%2F%2Flocalhost%3A9005

Waiting for authentication...

+  Success! Logged in as meriammartadinata@gmail.com
PS D:\SEMESTER 4\Pemrograman-Mobile\WEEK 8-API\src>
```

- Aktivasi FlutterFire CLI secara global untuk sinkronisasi Flutter dengan Firebase.

```
PS D:\SEMESTER 4\Pemrograman-Mobile\WEEK 8-API\src> dart pub global activate flutterfire_cli
+ checked_yaml 2.0.4
+ ci 0.1.0
+ cli_util 0.4.2 (0.5.1 available)
+ clock 1.1.2
+ collection 1.19.1
+ dart_console 1.2.0 (4.1.4 available)
+ deep_pick 1.1.0
+ ffi 2.2.0
+ file 7.0.1
+ flutterfire_cli 1.3.2
+ http 1.6.0
+ http_parser 4.1.2
+ interact 2.2.0
+ intl 0.18.1 (0.20.2 available)
+ json_annotation 4.11.0
+ meta 1.18.2
+ path 1.9.1
+ petitparser 7.0.2
+ platform 3.1.6
+ process 5.0.5
+ pub_semver 2.2.0
+ pub_updater 0.5.0
+ pubspec_parse 1.5.0
+ source_span 1.10.2
+ string_scanner 1.4.1
+ term_glyph 1.2.2
+ tint 2.0.1
+ typed_data 1.4.0
+ web 1.1.1
+ win32 5.15.0 (6.2.0 available)
+ xml 6.6.1 (7.0.1 available)
+ yaml 3.1.3
```

- Konfigurasi otomatis file Firebase Options ke dalam project Flutter melalui perintah flutterfire configure

```
PROBLEMS 4   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   POSTMAN CONSOLE

PS D:\SEMESTER 4\Pemrograman-Mobile\WEEK 8-API\src\aplikasi_cuaca> flutterfire configure
i Found 1 Firebase projects.
? Select a Firebase project to configure your Flutter application with ›
❯ aplikasi-cuaca-dd7ba (Aplikasi cuaca)
  <create a new project>
```

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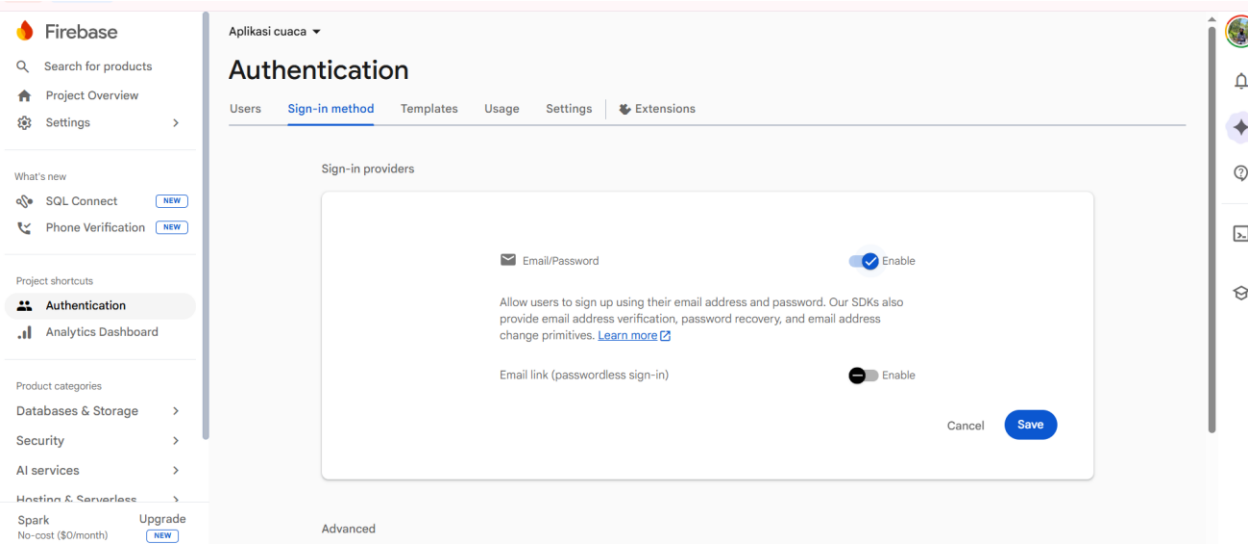
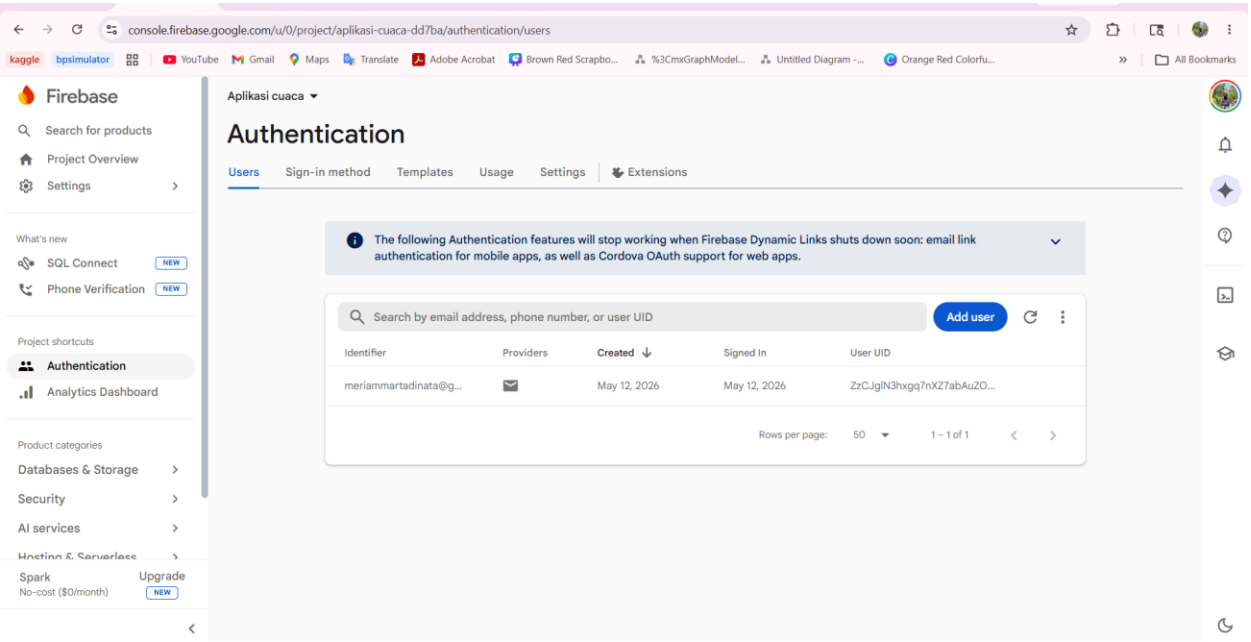
- Aktivasi metode *Sign-in* menggunakan Email/Password pada Firebase Authentication Console.



- Daftar pengguna yang berhasil terdaftar secara *real-time* di database Firebase Authentication.



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- Impelementasi Halaman Login dan Register

> login

lib > screens > login_screen.dart > _LoginScreenState

```dart
import 'package:flutter/material.dart';
import 'register_screen.dart';
import 'package:firebase_auth/firebase_auth.dart';
import 'home_screen.dart';

class LoginScreen extends StatefulWidget {
  const LoginScreen({super.key});

  @override
  State<LoginScreen> createState() => _LoginScreenState();
}

class _LoginScreenState extends State<LoginScreen> {
  final TextEditingController _emailController = TextEditingController();
  final TextEditingController _passwordController = TextEditingController();

  // --- FUNGSI LOGIN FIREBASE ---
  Future<void> _login() async {
    try {
      // Mengirim data ke Firebase untuk dicocokkan
      await FirebaseAuth.instance.signInWithEmailAndPassword(
        email: _emailController.text.trim(),
        password: _passwordController.text.trim(),
      );

      if (mounted) {
        Navigator.pushReplacement(
          context,
          MaterialPageRoute(builder: (context) => const HomeScreen()),
        );
      }
    } on FirebaseAuthException catch (e) {
      // Menangkap pesan error dari Firebase
      ScaffoldMessenger.of(context).showSnackBar(    Don't use 'BuildContext's across async gaps.Try rewriting the code to not use the 'Buil
        SnackBar(
          content: Text(e.message ?? "Terjadi kesalahan saat login"),
          backgroundColor: Colors.redAccent,
        ), // SnackBar
      );
    }
  }

  // --- FUNGSI REGISTER FIREBASE ---
  Future<void> _register() async {    The declaration '_register' isn't referenced.Try removing the declaration of '_register'.
```

```dart
    // Membuat akun baru di database Firebase
    await FirebaseAuth.instance.createUserWithEmailAndPassword(
      email: _emailController.text.trim(),
      password: _passwordController.text.trim(),
    );

    if (mounted) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(
          content: Text("Akun berhasil dibuat! Silakan klik tombol LOGIN."),
          backgroundColor: Colors.green,
        ), // SnackBar
      );
    }
  } on FirebaseAuthException catch (e) {
    ScaffoldMessenger.of(context).showSnackBar(    Don't use 'BuildContext's across async gaps.Try rewriting the code to not use the 'Buil
      SnackBar(
        content: Text(e.message ?? "Gagal membuat akun"),
        backgroundColor: Colors.redAccent,
      ), // SnackBar
    );
  }
}

@override
void dispose() {
  _emailController.dispose();
  _passwordController.dispose();
  super.dispose();
```

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> Register

```
main.dart U | api_config.dart U | app_colors.dart U | weather_model.dart U | auth_service.dart U | login_screen.dart 1, U | register_screen.dart U | home_screen.dart U | forecast_card.dart U | detail
lib > screens > register_screen.dart > RegisterScreen > RegisterScreen
import 'package:flutter/material.dart';
import 'package:firebase_auth/firebase_auth.dart';

class RegisterScreen extends StatefulWidget {
  const RegisterScreen({super.key});

  @override
  State<RegisterScreen> createState() => _RegisterScreenState();
}

class _RegisterScreenState extends State<RegisterScreen> {
  final TextEditingController _emailController = TextEditingController();
  final TextEditingController _passwordController = TextEditingController();
  final TextEditingController _confirmPasswordController = TextEditingController();

  Future<void> _register() async {
    // Validasi apakah password sama
    if (_passwordController.text != _confirmPasswordController.text) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(content: Text("Password tidak cocok!"), backgroundColor: Colors.orange),
      );
      return;
    }

    try {
      await FirebaseAuth.instance.createUserWithEmailAndPassword(
        email: _emailController.text.trim(),
        password: _passwordController.text.trim(),
      );

      if (mounted) {
        ScaffoldMessenger.of(context).showSnackBar(
          const SnackBar(content: Text("Akun berhasil dibuat! Silakan Login."), backgroundColor: Colors.green),
        );
        Navigator.pop(context); // Kembali ke halaman Login
      }
    } on FirebaseAuthException catch (e) {
      ScaffoldMessenger.of(context).showSnackBar(    Don't use 'BuildContext's across async gaps.Try rewriting the code to not use the 'BuildContext', or guard the use with a 'mounted' check.
        SnackBar(content: Text(e.message ?? "Gagal mendaftar"), backgroundColor: Colors.redAccent),
      );
    }
  }

```

```
// --- REGISTER CARD ---
Container(
  margin: const EdgeInsets.symmetric(horizontal: 24),
  padding: const EdgeInsets.all(20),
  decoration: BoxDecoration(
    color: Colors.white,
    borderRadius: BorderRadius.circular(25),
    boxShadow: [
      BoxShadow(color: Colors.black12, blurRadius: 15, offset: Offset(0, 5))
    ],
  ), // BoxDecoration
  child: Column(
    children: [
      TextField(
        controller: _emailController,
        decoration: _inputStyle("Email", Icons.email_outlined),
      ), // TextField
      const SizedBox(height: 12),
      TextField(
        controller: _passwordController,
        obscureText: true,
        decoration: _inputStyle("Password", Icons.lock_outline),
      ), // TextField
      const SizedBox(height: 12),
      TextField(
        controller: _confirmPasswordController,
        obscureText: true,
        decoration: _inputStyle("Confirm Password", Icons.lock_reset_outlined),
      ), // TextField
      const SizedBox(height: 20),
      SizedBox(
        width: double.infinity,
        child: ElevatedButton(
          style: ElevatedButton.styleFrom(
            backgroundColor: const Color(0xFF5A9CFF),
            padding: const EdgeInsets.symmetric(vertical: 14),
            shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(12)),
            elevation: 0,
```

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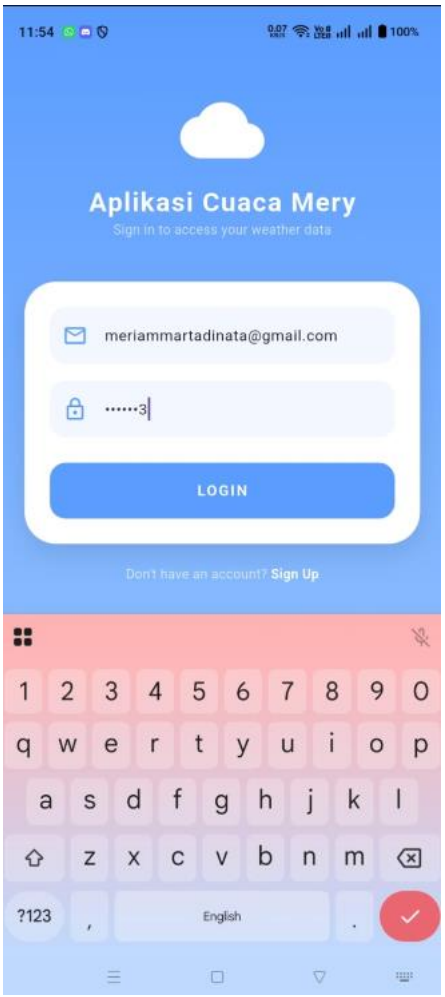

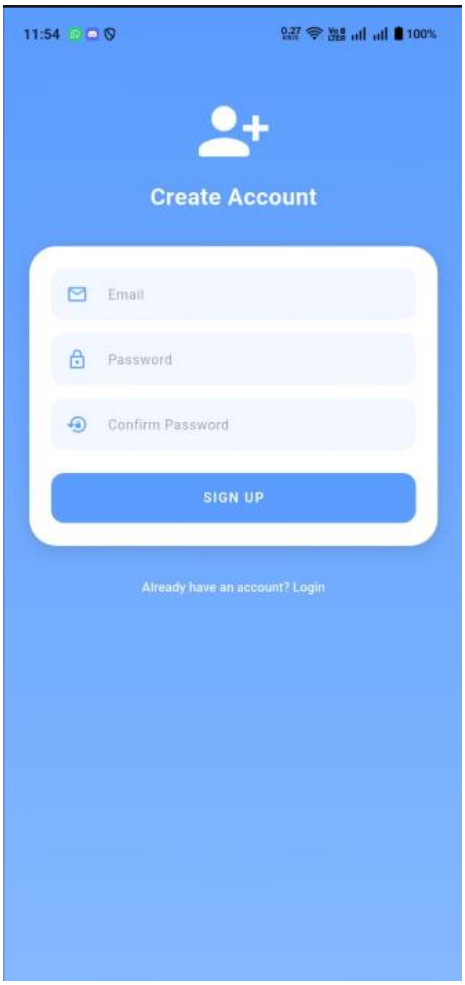

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2. Uji Coba API Cuaca (Open-Meteo)

- Models /weather/model.dart

File models/weather_model.dart adalah cetakan (blueprint) yang berfungsi untuk memetakan data JSON yang sangat kompleks dari API menjadi objek yang mudah dibaca oleh Flutter.

```dart
class WeatherModel {
  final double temperature;
  final double windSpeed;
  final int weatherCode;
  final int humidity;
  final List<double> maxTemp;

  WeatherModel({
    required this.temperature,
    required this.windSpeed,
    required this.weatherCode,
    required this.humidity,
    required this.maxTemp,
  });

  factory WeatherModel.fromJson(Map<String, dynamic> json) {
    // Ambil data 'current' (untuk kelembapan)
    final current = json['current'];
    // Ambil data 'current_weather' (untuk suhu & angin)
    final currentWeather = json['current_weather'];
    // Ambil data 'daily' (untuk suhu harian)
    final daily = json['daily'];

    // Pengaman List Suhu Harian
    List<double> temps = [0.0, 0.0, 0.0];
    if (daily != null && daily['temperature_2m_max'] != null) {
      temps = List<double>.from(daily['temperature_2m_max'].map((x) => x.toDouble()));
    }

    return WeatherModel(
      // Jika suhu kosong, default ke 0.0
      temperature: (currentWeather?['temperature'] ?? 0.0).toDouble(),
      // Jika kecepatan angin kosong, default ke 0.0
      windSpeed: (currentWeather?['windspeed'] ?? 0.0).toDouble(),
      // Jika kode cuaca kosong, default ke 0
      weatherCode: (currentWeather?['weathercode'] ?? 0).toInt(),
      // Jika kelembapan kosong, default ke 0
      humidity: (current?['relative_humidity_2m'] ?? 0).toInt(),
      maxTemp: temps,
    );
  }
}
```

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- Services/weather_api. Dart

Kode yang menghubungkan aplikasi Flutter dengan server Open-Meteo di internet. Di sini menggunakan *library* bernama Dio untuk mengirim permintaan (HTTP Request).

lib > services > weather_api.dart > WeatherApiClient

```dart
import 'package:dio/dio.dart';
import '../models/weather_model.dart';

class WeatherApiClient {
 Future<WeatherModel> fetchWeather(double lat, double lon) async {
  final String url =
    "https://api.open-meteo.com/v1/forecast?latitude=$lat&longitude=$lon&current=relative_humidity_2m&current_weather=true&daily=temperature_2m_max&past_days=1";

  try {
    Response response = await Dio().get(url);
    if (response.statusCode == 200) {
      return WeatherModel.fromJson(response.data);
    } else {
      throw Exception("Gagal memuat data dari server");
    }
  } catch (e) {
    throw Exception("Masalah Koneksi: $e");
  }
}
  Future<Map<String, dynamic>> getCoordinates(String cityName) async {
    final String url = "https://geocoding-api.open-meteo.com/v1/search?name=$cityName&count=1&language=en&format=json";
    try {
      Response response = await Dio().get(url);
      if (response.data['results'] != null) {
        final res = response.data['results'][0];
        return {"lat": res['latitude'], "lon": res['longitude'], "name": res['name']};
      }
      throw Exception("Kota tidak ditemukan");
    } catch (e) {
      throw Exception("Gagal mencari lokasi");
    }
  }
}
```

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-Screen/homescreen.dart

Menampilkan tampilan aplikasi

lib > screens > home_screen.dart > _HomeScreenState

```dart
import 'package:flutter/material.dart';
import 'package:intl/intl.dart';
import '../services/weather_api.dart';
import '../models/weather_model.dart';

class HomeScreen extends StatefulWidget {
  const HomeScreen({super.key});

  @override
  State<HomeScreen> createState() => _HomeScreenState();
}

class _HomeScreenState extends State<HomeScreen> {
  final WeatherApiClient _weatherService = WeatherApiClient();
  final TextEditingController _searchController = TextEditingController();

  String _currentCity = "Malang";
  double _lat = -7.98;
  double _lon = 112.63;
  int _selectedTab = 1;

  late Future<WeatherModel> _weatherFuture;

  @override
  void initState() {
    super.initState();
    _weatherFuture = _weatherService.fetchWeather(_lat, _lon);
  }

  void _searchCity() async {
    if (_searchController.text.isEmpty) return;
    try {
      final coords = await _weatherService.getCoordinates(_searchController.text);
      setState(() {
        _lat = coords['lat'];
        _lon = coords['lon'];
        _currentCity = coords['name'];
        _selectedTab = 1;
        _weatherFuture = _weatherService.fetchWeather(_lat, _lon);
      });
      _searchController.clear();
      FocusScope.of(context).unfocus();    Don't use 'BuildContext's across async gaps.⏎Try rewriting the code to not use the 'BuildContext', or guard the use with a 'mou
    } catch (e) {
      ScaffoldMessenger.of(context).showSnackBar(    Don't use 'BuildContext's across async gaps.⏎Try rewriting the code to not use the 'BuildContext', or guard the use w
        const SnackBar(content: Text("Kota tidak ditemukan")),
```

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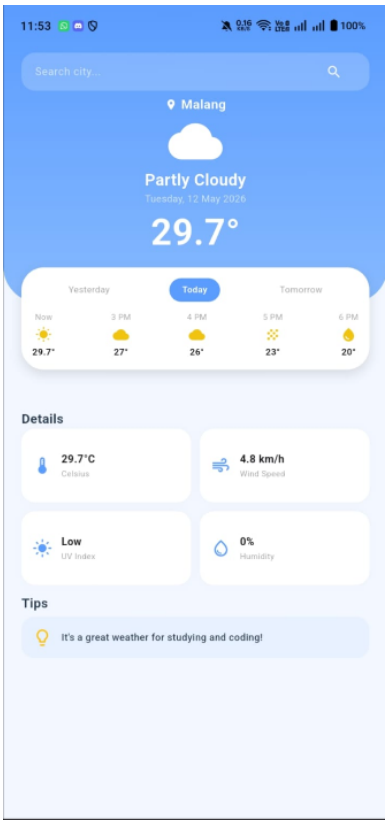

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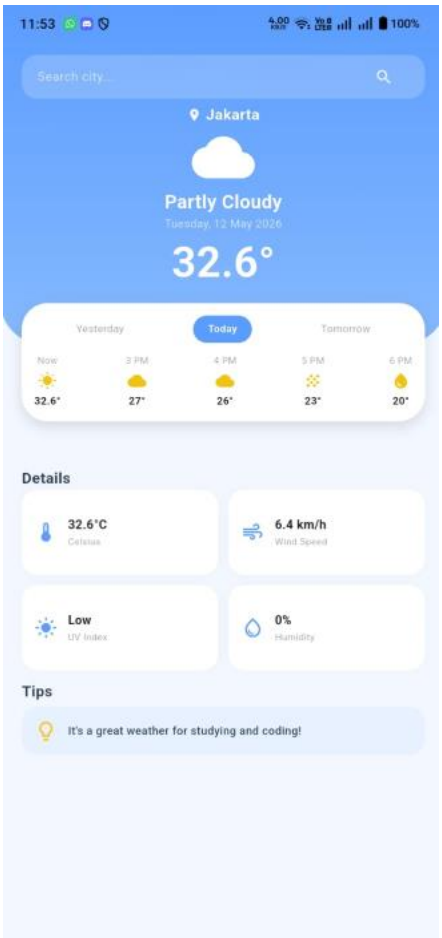
Kita bisa mencari prakiraan cuaca sesuai dengan kota/negara yang kita inginkan